

# Sharpness-Aware Minimization

**Input:** Training set  $\mathcal{S} \triangleq \cup_{i=1}^n \{(\mathbf{x}_i, \mathbf{y}_i)\}$ , Loss function  $l : \mathcal{W} \times \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}_+$ , Batch size  $b$ , Step size  $\eta > 0$ , Neighborhood size  $\rho > 0$ .

**Output:** Model trained with SAM

Initialize weights  $\mathbf{w}_0$ ,  $t = 0$ ;

**while not converged do**

    Sample batch  $\mathcal{B} = \{(\mathbf{x}_1, \mathbf{y}_1), \dots (\mathbf{x}_b, \mathbf{y}_b)\}$ ;

    Compute gradient  $\nabla_{\mathbf{w}} L_{\mathcal{B}}(\mathbf{w})$  of the batch's training loss;

    Compute  $\hat{\epsilon}(\mathbf{w})$  per equation 2;

    Compute gradient approximation for the SAM objective

        (equation 3):  $\mathbf{g} = \nabla_{\mathbf{w}} L_{\mathcal{B}}(\mathbf{w})|_{\mathbf{w} + \hat{\epsilon}(\mathbf{w})}$ ;

    Update weights:  $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \mathbf{g}$ ;

$t = t + 1$ ;

**end**

**return**  $\mathbf{w}_t$

**Algorithm 1:** SAM algorithm

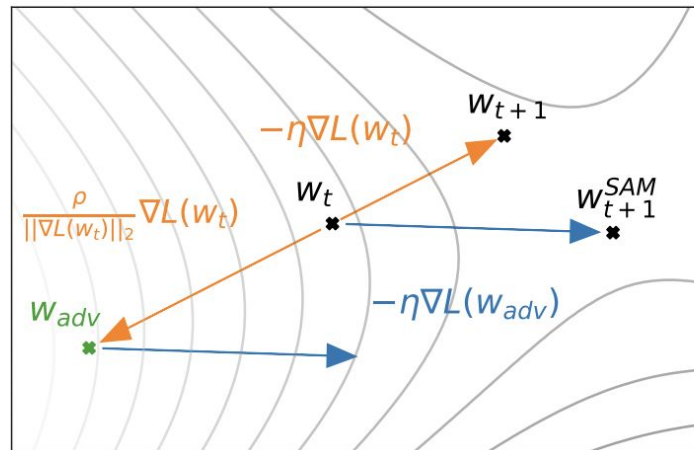


Figure 2: Schematic of the SAM parameter update.

$$\mathbb{E}_{\mathbf{w} \sim \mathcal{Q}}[L_{\mathcal{D}}(\mathbf{w})] \leq \mathbb{E}_{\mathbf{w} \sim \mathcal{Q}}[L_{\mathcal{S}}(\mathbf{w})] + \sqrt{\frac{KL(\mathcal{Q}||\mathcal{P}) + \log \frac{n}{\delta}}{2(n-1)}} \quad (5)$$

## PAC-Bayes bounds

**Theorem 2.** *For any  $\rho > 0$  and any distribution  $\mathcal{D}$ , with probability  $1 - \delta$  over the choice of the training set  $\mathcal{S} \sim \mathcal{D}$ ,*

$$L_{\mathcal{D}}(\mathbf{w}) \leq \max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} L_{\mathcal{S}}(\mathbf{w} + \boldsymbol{\epsilon}) + \sqrt{\frac{k \log \left( 1 + \frac{\|\mathbf{w}\|_2^2}{\rho^2} \left( 1 + \sqrt{\frac{\log(n)}{k}} \right)^2 \right) + 4 \log \frac{n}{\delta} + \tilde{O}(1)}{n-1}} \quad (4)$$

where  $n = |\mathcal{S}|$ ,  $k$  is the number of parameters and we assumed  $L_{\mathcal{D}}(\mathbf{w}) \leq \mathbb{E}_{\boldsymbol{\epsilon}_i \sim \mathcal{N}(0, \rho)}[L_{\mathcal{D}}(\mathbf{w} + \boldsymbol{\epsilon})]$ .

**Theorem (stated informally) 1.** *For any  $\rho > 0$ , with high probability over training set  $\mathcal{S}$  generated from distribution  $\mathcal{D}$ ,*

$$L_{\mathcal{D}}(\mathbf{w}) \leq \max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} L_{\mathcal{S}}(\mathbf{w} + \boldsymbol{\epsilon}) + h(\|\mathbf{w}\|_2^2 / \rho^2),$$

*where  $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  is a strictly increasing function (under some technical conditions on  $L_{\mathcal{D}}(\mathbf{w})$ ).*

Sharpness

Loss

Regulator

$$\left[ \max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} L_{\mathcal{S}}(\mathbf{w} + \boldsymbol{\epsilon}) - L_{\mathcal{S}}(\mathbf{w}) \right] + L_{\mathcal{S}}(\mathbf{w}) + h(\|\mathbf{w}\|_2^2 / \rho^2).$$

$$L_{\mathcal{D}}(\boldsymbol{w}) \leq \max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} L_{\mathcal{S}}(\boldsymbol{w} + \boldsymbol{\epsilon}) + h(\|\boldsymbol{w}\|_2^2 / \rho^2),$$

$$\min_{\boldsymbol{w}} L_{\mathcal{S}}^{SAM}(\boldsymbol{w}) + \lambda \|\boldsymbol{w}\|_2^2 \quad \text{where} \quad L_{\mathcal{S}}^{SAM}(\boldsymbol{w}) \triangleq \max_{\|\boldsymbol{\epsilon}\|_p \leq \rho} L_{\mathcal{S}}(\boldsymbol{w} + \boldsymbol{\epsilon}),$$

$$\boldsymbol{\epsilon}^*(\boldsymbol{w}) \triangleq \arg \max_{\|\boldsymbol{\epsilon}\|_p \leq \rho} L_{\mathcal{S}}(\boldsymbol{w} + \boldsymbol{\epsilon}) \approx \arg \max_{\|\boldsymbol{\epsilon}\|_p \leq \rho} L_{\mathcal{S}}(\boldsymbol{w}) + \boldsymbol{\epsilon}^T \nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w}) = \arg \max_{\|\boldsymbol{\epsilon}\|_p \leq \rho} \boldsymbol{\epsilon}^T \nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w}).$$

$$\hat{\boldsymbol{\epsilon}}(\boldsymbol{w}) = \rho \operatorname{sign}(\nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w})) |\nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w})|^{q-1} / \left( \|\nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w})\|_q^q \right)^{1/p}$$

$$\begin{aligned}\nabla_{\boldsymbol{w}} L_{\mathcal{S}}^{SAM}(\boldsymbol{w}) &\approx \nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w} + \hat{\boldsymbol{\epsilon}}(\boldsymbol{w})) = \frac{d(\boldsymbol{w} + \hat{\boldsymbol{\epsilon}}(\boldsymbol{w}))}{d\boldsymbol{w}} \nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w})|_{\boldsymbol{w} + \hat{\boldsymbol{\epsilon}}(\boldsymbol{w})} \\ &= \nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w})|_{\boldsymbol{w} + \hat{\boldsymbol{\epsilon}}(\boldsymbol{w})} + \frac{d\hat{\boldsymbol{\epsilon}}(\boldsymbol{w})}{d\boldsymbol{w}} \nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w})|_{\boldsymbol{w} + \hat{\boldsymbol{\epsilon}}(\boldsymbol{w})}.\end{aligned}$$

$$\nabla_{\boldsymbol{w}} L_{\mathcal{S}}^{SAM}(\boldsymbol{w}) \approx \nabla_{\boldsymbol{w}} L_{\mathcal{S}}(\boldsymbol{w})|_{\boldsymbol{w} + \hat{\boldsymbol{\epsilon}}(\boldsymbol{w})}.$$

# Неявная реализация

## 2.1 GRADIENT REGULARIZATION

We consider GR (Barrett & Dherin, 2021; Smith et al., 2021), wherein the squared L2 norm of the gradient is explicitly added to the original loss  $\mathcal{L}(\theta)$  as follows:

$$\tilde{\mathcal{L}}(\theta) = \mathcal{L}(\theta) + \frac{\gamma}{2}R(\theta), \quad R(\theta) = \|\nabla\mathcal{L}(\theta)\|^2, \quad (1)$$

where  $\|\cdot\|$  denotes the Euclidean norm and  $\gamma > 0$  is a constant regularization coefficient. We abbreviate the derivative with respect to the parameters  $\nabla_{\theta}$  by  $\nabla$ . Its gradient descent is given by

$$\theta_{t+1} = \theta_t - \eta \nabla \tilde{\mathcal{L}}(\theta_t) \quad (2)$$

for time step  $t = 0, 1, \dots$  and learning rate  $\eta > 0$ . While previous studies have reported that explicitly adding a GR term empirically improves generalization performance, its algorithms and implementations have not been discussed in much detail.

# Difference Scheme

We update the GR term by

$$\Delta R_F(\varepsilon) = \frac{\nabla \mathcal{L}(\theta_t + \varepsilon \nabla \mathcal{L}(\theta_t)) - \nabla \mathcal{L}(\theta_t)}{\varepsilon} \quad (\text{F-GR}). \quad (3)$$

We refer to this gradient as *Forward finite-difference GR (F-GR)*. Because the gradient  $\nabla \mathcal{L}(\theta_t)$  is computed for the original loss, the finite difference (3) requires only one additional gradient computation  $\nabla \mathcal{L}(\theta')$ . The order of the computation time is only double that of the usual gradient descent. The finite-difference method also has a backward computation:

$$\Delta R_B(\varepsilon) = \frac{\nabla \mathcal{L}(\theta_t) - \nabla \mathcal{L}(\theta_t - \varepsilon \nabla \mathcal{L}(\theta_t))}{\varepsilon} \quad (\text{B-GR}). \quad (4)$$

# FGR vs SAM

$$\nabla \mathcal{L}(\theta) + \frac{\gamma_t}{\varepsilon_t}(\nabla \mathcal{L}(\theta') - \nabla \mathcal{L}(\theta)) = \nabla \mathcal{L}(\theta'). \quad (19)$$

A similar equivalence has been pointed out in [Zhao et al. \(2022\)](#) which supposes a non-squared gradient norm and  $\varepsilon_t = \rho / \|\nabla \mathcal{L}(\theta_t)\|$  naturally appears. Suppose the SAM update without the gradient normalization for simplicity, that is,  $\varepsilon_t = \rho$ . This simplified SAM update was analyzed on DLNs in [Andriushchenko & Flammarion \(2022\)](#). We can recover their expression of  $\alpha$  by setting a sufficiently small  $\gamma = \varepsilon$  and neglecting  $c_1$  and  $c_2$  terms in Theorem 4.3.



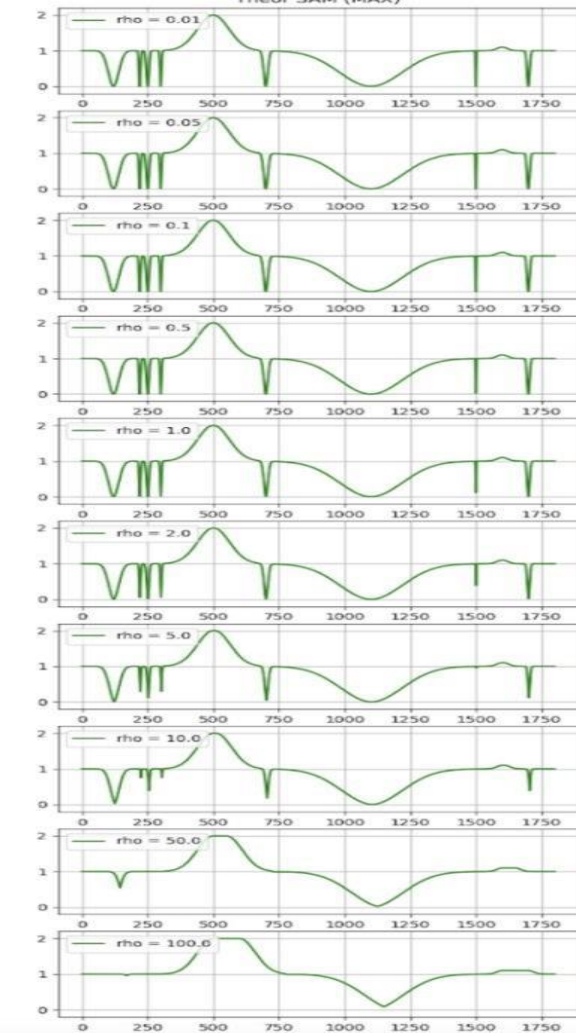
# Экстраградиентный метод (Korpelevich, 1976)

Therefore, we consider operator  $F_{\text{EG},\gamma} = F(\text{Id} - \gamma F)$  defining the update of EG:

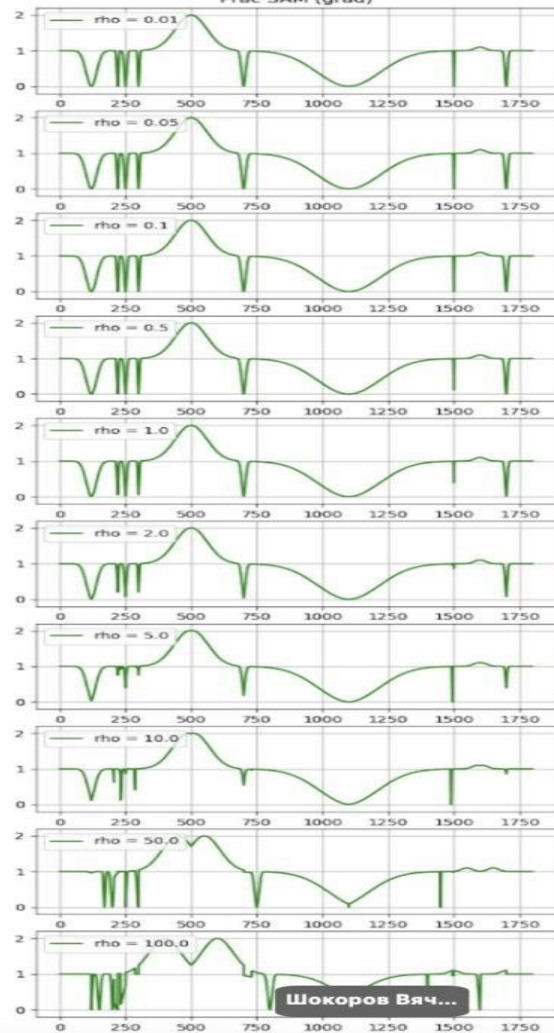
$$x^{k+1} = x^k - \underbrace{\gamma F(x^k - \gamma F(x^k))}_{F_{\text{EG},\gamma}(x^k)}. \quad (\text{EG})$$

Extragradient method (EG) (Korpelevich, 1976) is one of the most popular methods for solving saddle point and variational inequalities problems (VIP).

Theor SAM (MAX)



Prac SAM (grad)



Grad norm square penalization

